Optimal Margin Classifier

$$\min_{\gamma, w, b} \gamma \quad \text{s.t} \quad y^{(i)}(w^T x^{(i)} + b) \geq \gamma \quad \Rightarrow \quad \min_{\hat{\gamma}, w, b} \frac{\hat{\gamma}}{\|w\|} \quad \text{s.t} \quad y^{(i)}(w^T x^{(i)} + b) \geq \hat{\gamma} \quad \Rightarrow \quad \min_{w, b} \frac{1}{\|w\|} \quad \text{s.t} \quad y^{(i)}(w^T x^{(i)} + b) \geq 1$$

Lagrangian $\min_w f(w)$ s.t $h_i(w) = 0$ then $\mathcal{L}(w, b) = f(w) + \sum_i \beta_i h_i(w)$

$$\frac{\partial \mathcal{L}}{\partial w_i} = 0 \qquad \frac{\partial \mathcal{L}}{\partial \beta_i} = 0$$

Generalized Lagrangian $\min_w f(w)$ s.t $g_i(w) \leq 0$, $h_i(w) = 0$ then $\mathcal{L}(w, \alpha, \beta) = f(w) + \sum_i \alpha_i g_i(w) + \sum_i \beta_i h_i(w)$

Primal Constraints. then $\theta_P(w) = \max_{\alpha, \beta, \alpha \geq 0} \left\{ f(w) + \sum_i \alpha_i g_i(w) + \sum_i \beta_i h_i(w) \right\}$

if $g_i(w) \geq 0$ or $h_i(w) \neq 0$ $= \infty$

otherwise $\theta_P(w) = f(w)$

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primal problem $p^* = \min_w \max_{\alpha, \beta : \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta)$

$\Rightarrow \quad d^* \leq p^*$

dual problem $d^* = \max_{\alpha, \beta : \alpha_i \geq 0} \min_w \mathcal{L}(w, \alpha, \beta)$

(i) f, g_i convex

(ii) h_i are affine : $h_i(w) = a_i^T w + b_i$

(iii) g_i are strictly feasible : $\exists w$ s.t $\forall g_i(w) < 0$

$\Rightarrow \quad p^* = d^* = \mathcal{L}(w^*, \alpha^*, \beta^*)$

KKT conditions

(i) $\frac{\partial}{\partial w_i} \mathcal{L}(w^*, \alpha^*, \beta^*) = 0$

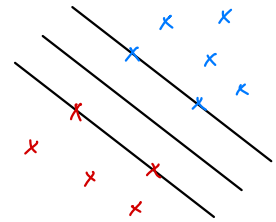
(ii) $\frac{\partial}{\partial \beta_i} \mathcal{L}(w^*, \alpha^*, \beta^*) = 0$

(iii) $\alpha_i^* g_i(w^*) = 0 \rightsquigarrow \alpha_i^* > 0$ then $g_i(w^*) = 0$

(iv) $g_i(w^*) \leq 0$

(v) $\alpha^* \geq 0$

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{s.t} \quad g_i(w) = 1 - y^{(i)}(w^T x^{(i)} + b) \le 0$$



- $\mathcal{L}(w, \alpha, \beta) = \frac{1}{2}\|w\|^2 - \sum_i \alpha_i \left[y^{(i)}(w^T x^{(i)} + b) - 1 \right]$

- $\frac{\partial \mathcal{L}}{\partial w} = 0$ implies $w = \sum_i \alpha_i y^{(i)} x^{(i)}$

- $\frac{\partial \mathcal{L}}{\partial b} = 0$ implies $0 = \sum_i \alpha_i y^{(i)}$

$$\Rightarrow \mathcal{L}(w, \alpha, \beta) = \sum_i \alpha_i - \frac{1}{2} w^T w$$

$$\Rightarrow d^* = \max_{\alpha} \left(\sum_i \alpha_i - \frac{1}{2} w^T w \right) \quad \text{s.t} \quad \alpha_i \ge 0 \quad w = \sum_i \alpha_i y^{(i)} x^{(i)}$$

$$0 = \sum_i \alpha_i y^{(i)}$$

Suppose $\alpha_i > 0$

Note.

$$w^T x + b = \sum_i \alpha_i y^{(i)} \langle x^{(i)}, x \rangle$$

(i) $g_i(w) = 0$ i.e support vector

(ii) $y^{(i)}(w^{*T} x^{(i)} + b) = 1$

$$w^{*T} x^{(i)} + b = y^{(i)}$$

$$\Rightarrow b^* = -\frac{1}{2} \cdot \left\{ \max_{i : y_i = -1} w^T x^{(i)} + \min_{i : y_i = 1} w^T x^{(i)} \right\}$$

$$\min_{w,b,\xi} \quad \frac{1}{2}\|w\|^2 + C\sum_i \xi_i \quad \text{s.t} \quad y^{(i)}(w^T x^{(i)} + b) \geq \underbrace{1-\xi_i}_{\text{permitted to have margin } < 1} \qquad \xi_i \geq 0$$

$$\mathcal{L} = \frac{1}{2} w^T w + C\sum_i \xi_i - \sum_i \alpha_i \left\{ y^{i}(w^T x^{(i)} + b) - 1 + \xi_i \right\} - \sum_i r_i \xi_i$$

- $\frac{\partial \mathcal{L}}{\partial \xi_i} = 0$ implies $C = \alpha_i + r_i$ i.e $0 \leq \alpha_i \leq C$
- $\frac{\partial \mathcal{L}}{\partial w} = 0$ implies $w = \sum_i \alpha_i y^{(i)} x^{(i)}$
- $\frac{\partial \mathcal{L}}{\partial b} = 0$ implies $0 = \sum_i \alpha_i y^{(i)}$

$$\Rightarrow \quad \max_{\alpha} \quad \sum \alpha_i - \frac{1}{2} w^T w \quad \text{s.t} \quad 0 \leq \alpha_i \leq C$$

$$w = \sum_i \alpha_i y^{(i)} x^{(i)}$$

$$0 = \sum_i \alpha_i y^{(i)}$$

$\alpha_i = 0$ then $y^{(i)}(w^T x^{(i)} + b) \geq 1$

$0 < \alpha_i < C$ then $y^{(i)}(w^T x^{(i)} + b) = 1$

$\alpha_i = C$ then $y^{(i)}(w^T x^{(i)} + b) \leq 1$

$\because$

(i) $y^{(i)}(w^T x^{(i)} + b) \geq 1 - \xi_i$

$\xi_i \geq 0$

(ii) $\alpha_i \left(y^{(i)}(w^T x^{(i)} + b) - 1 + \xi_i \right) = 0$

$r_i \xi_i = 0$

(iii) $C = \alpha_i + r_i$

Coordinate Ascent

: $a_i = \arg\max_{\hat{\alpha}_i} W(\alpha_1 \dots \alpha_{i-1}\ \hat{\alpha}_i\ \alpha_{i+1} \dots \alpha_n)$

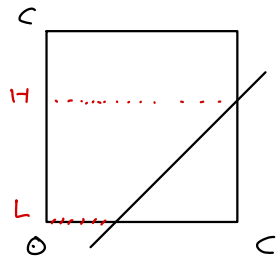
$\rightarrow\leftarrow$ $\sum_i a_i y^{(i)} = 0$ $\qquad \alpha_1 y^{(1)} = -\sum_{i=2}^{n} \alpha_i y^{(i)}$

SMO Algorithm.

- 직선 $\alpha_1 y^{(1)} + \alpha_2 y^{(2)} = -\sum_{i=3}^{n} \alpha_i y^{(i)}$

$= \xi$

with $0 \le \alpha_1, \alpha_2 \le C$



- $\alpha_1 = y^{(1)}(\xi - \alpha_2 y^{(2)})$

- $\alpha_2 = \arg\max_{L \le \alpha_2 \le H} W(y^{(1)}(\xi - \alpha_2 y^{(1)}),\ \alpha_2 \dots)$